

Java Statement-Based Mini Projects

10 classroom problems covering Variables, Data Types, Scanner, Operators, if, if-else, else-if, logical operators, increment/decrement and nested if.

Student Rule: Read the situation first. Decide the logic yourself. Do not start by writing syntax. Identify the inputs, calculations, conditions and outputs.

1. Student Result & Scholarship System ■

A college wants a simple program to evaluate a student's academic performance. Take the student's name, marks in three subjects and attendance percentage as input. Calculate total marks and percentage. If the percentage is below 40, display **Fail**. Otherwise, assign a grade: 90+ = A, 75–89 = B, 60–74 = C, 40–59 = D. If the student has at least 80% attendance and percentage is 75 or above, display **Scholarship Eligible**; otherwise display **No Scholarship**.

Inputs: Name • Marks in 3 subjects • Attendance %

Think: Use arithmetic operators for total/percentage, else-if for grade, and && for scholarship eligibility.

2. ATM Withdrawal System ■

Create a basic ATM program. Take account balance, PIN and withdrawal amount as input. First check whether the PIN is correct. If it is correct, check whether the withdrawal amount is positive and does not exceed the balance. If valid, deduct the amount from the balance and display the remaining balance. Also display a warning if the remaining balance becomes less than ■1,000. If the PIN is wrong, display **Invalid PIN**.

Inputs: Current balance • PIN • Withdrawal amount

Think: Practice nested if, relational operators, arithmetic operators and assignment operators.

3. Online Shopping Checkout ■

An online store asks for product price, quantity, membership status and coupon code. Calculate the subtotal. If the subtotal is ■2,000 or more, give a 10% discount. Premium members get an additional 5% discount. If the final amount is ■999 or more, delivery is free; otherwise add ■80 delivery charges. Display the final payable amount.

Inputs: Product price • Quantity • Premium member: yes/no • Coupon code

Think: Use arithmetic calculations, if/else, nested conditions and logical operators.

4. Electricity Bill Calculator ■

A household electricity board wants to calculate a monthly bill. Take units consumed as input. Calculate the bill using these slabs: first 100 units at ■5/unit, next 100 units at ■7/unit, and units above 200 at ■10/unit. If the calculated bill is above ■2,500, add a 5% surcharge. Display the units, base bill, surcharge and final bill.

Inputs: Units consumed

Think: Focus on else-if/slab logic, arithmetic operators and updating a running bill.

5. Food Delivery Order ■

A food delivery app takes food price, quantity, distance in km and whether the customer has membership. Calculate the food total. If the order total is ■500 or more, delivery is free; otherwise charge ■50. If the customer is a member, give ■100 off when the order total is ■1,000 or more. If the delivery distance is more than 10 km, add ■30 extra. Display the final amount.

Inputs: Food price • Quantity • Distance • Membership yes/no

Think: Combine arithmetic, if-else, multiple conditions and logical operators.

6. Funny College Attendance Rescue ■

A student enters attendance percentage, number of assignments completed and whether they have submitted the project. The college rule is: attendance of 75% or more is required. If attendance is below 75%, display **Extra Classes Required** ■. If attendance is sufficient, check the project submission. If the project is submitted and at least 80% of assignments are completed, display **Enjoy Your Semester** ■. Otherwise display **Complete Your Work First!**.

Inputs: Attendance % • Assignments completed • Project submitted: yes/no

Think: Use nested if and &&. Add a funny output message, but keep the logic correct.

7. Employee Salary & Bonus System ■

A company wants to calculate an employee's monthly salary. Take basic salary, years of experience and performance rating (1–5). If the rating is 5, give a 20% bonus; rating 4 gives 10%; rating 3 gives 5%; otherwise no bonus. Employees with more than 5 years of experience get an additional ■2,000. If the final salary is above ■50,000, display **Senior Compensation Band**; otherwise display **Standard Compensation Band**.

Inputs: Basic salary • Years of experience • Performance rating

Think: Use else-if for rating, arithmetic for bonus/salary, and conditions for experience and band.

8. Mobile Data Usage Alert ■

A mobile app tracks a user's monthly data usage. Take total data purchased and data used in GB. Calculate remaining data. If remaining data is 0 or below, display **Data Exhausted**. If remaining data is 1 GB or less, display **Low Data Alert**. If the user has used more than 80% of the plan, display **High Usage Warning**. Otherwise display **Usage Normal**.

Inputs: Total data plan • Data used

Think: Use percentage calculation, relational operators and else-if/multiple conditions.

9. Movie Ticket Booking ■

Create a cinema ticket system. Take customer's age, number of tickets and ticket price as input. Children below 12 receive 30% discount. Senior citizens aged 60 or above receive 20% discount. Everyone else pays the normal price. If the number of tickets is 5 or more, give an additional ■100 group discount. Display the original total, discount and final amount.

Inputs: Age • Number of tickets • Ticket price

Think: Use if-else, nested/multiple conditions, arithmetic operators and increment/decrement if needed.

10. Personal Budget Manager ■

Create a simple monthly budget checker. Take monthly income and three expense amounts as input. Calculate total expenses and remaining money. If expenses are greater than income, display **Budget Exceeded** and show the deficit. Otherwise calculate the saving percentage. If savings are 30% or more, display **Excellent Saving**; 20–29% = **Good Saving**; 10–19% = **Average Saving**; below 10% = **Needs Improvement**. Also increase a counter by 1 when the user stays within budget.

Inputs: Monthly income • Three expense amounts

Think: Use variables, arithmetic, else-if, comparison, assignment and ++.

Classroom Challenge Format

For every project, students should first write these five things in their notebook:

- 1. Input:** What information does the user enter?
- 2. Variables:** Which variables and data types are required?
- 3. Calculation:** What arithmetic must happen before the decision?
- 4. Condition:** Which if / else-if / nested-if / logical operator is needed?
- 5. Output:** What should the program display?

Bonus Rule

Do not tell students which concept to use initially. Let them identify it from the statement. After 5–10 minutes, discuss the possible logic as a class and then code.